# VoiceForAll

Bidirectional AI-Powered Sign Language Translation System

Sukhnaaz Kaur

BSc Artificial Intelligence and Computer Science

University of Birmingham

Supervisor: Martin

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*This document is a summary report. The full extended dissertation, including complete methodology, detailed evaluation, and all supporting materials, is available in the project GitLab repository and is the primary artifact intended for assessment.*

***Abstract*—VoiceForAll is a real-time American Sign Language recognition system designed to bridge the communication barrier between deaf signers and hearing individuals. The system uses MediaPipe Holistic to extract hand landmarks from a standard laptop camera, a Bidirectional LSTM model trained on the PopSign ASL v1.0 dataset [1] to classify signs, and a locally-running large language model with confidence-based filtering to convert recognised signs into grammatically correct English sentences, which are then spoken aloud via text-to-speech, running entirely offline with no specialist hardware or internet connection required. The system was developed iteratively across six versions, systematically testing different architectures and feature representations. The central finding of the project is that reducing input features from 225 landmarks to 63 hand-only landmarks produced a greater improvement in real-world performance than any architectural change, eliminating a persistent gap between laboratory test accuracy and live webcam performance that reached as high as 53 percentage points in earlier versions. The final system recognises 248 ASL signs with 72.82% test accuracy. Real-world accuracy reaches 87.5% on visually distinct signs, with an overall live accuracy of 65.4% across the full 248-sign vocabulary, a gap explained by a small subset of signs that are inherently indistinguishable without facial landmark data, and transparently documented in the evaluation. This work demonstrates that accessible, privacy-preserving sign language translation is achievable on standard consumer hardware, contributing towards more inclusive everyday communication for the deaf community.**

## I. Introduction and Motivation

The motivation for this project is personal. Growing up, I watched my cousin, who is deaf and non-speaking, sit alone at family gatherings because no one around him knew sign language. During a recent visit, I wanted to have a conversation with him but found myself reduced to guessing at what he was trying to tell me. That experience made one thing clear: the problem is not with the person who is deaf, it is with the tools available to bridge the communication gap. This experience is far from unique. According to the World Federation of the Deaf, over 70 million people worldwide use sign language as their primary means of communication [2], yet the vast majority of hearing people have no knowledge of it whatsoever. In the United Kingdom alone, an estimated 151,000 people use British Sign Language, of whom 87,000 are Deaf [3], [4]. The communication barrier this creates limits access to healthcare, education, employment, and everyday social interaction. A common assumption is that deaf individuals can simply resort to written language as an alternative, but for many, written English is a second language, sign languages have their own grammar, syntax, and sentence structure entirely distinct from English [7]. Professional interpreters remain the gold standard but are critically scarce: according to the NRCPD, only 908 registered sign language interpreters serve a community of over 87,000 deaf BSL signers [3], making them expensive, difficult to book, and unavailable for spontaneous everyday interactions. Existing digital tools partially address the problem but none solve it completely. Speech-to-text applications help deaf people understand hearing speakers but do nothing to help them be understood in return, while sign language translation tools either require specialist hardware, depend on an internet connection, or lack the accuracy needed for reliable real-world use [16]–[18]. VoiceForAll is a real-time sign language translation system designed to address this gap. Using only a standard laptop camera, the system detects hand movements using MediaPipe [5], recognises American Sign Language signs using a Bidirectional LSTM model [6] trained on the PopSign ASL v1.0 dataset [1], converts recognised signs into grammatically correct English sentences using a locally-running large language model [19], and speaks them aloud via text-to-speech all without any internet connection or specialist hardware. The full extended dissertation, including complete background literature, detailed version-by-version methodology, and full evaluation, is available in the project GitLab repository, which also contains all training scripts, preprocessing pipelines, trained model weights, and webcam inference scripts, with commit history tracking each version transition from V1 through V6.

## II. Background and Related Work

### A. Sign Language as a Linguistic System

Sign languages are complete, natural human languages with their own grammar, syntax, and linguistic structure entirely independent of spoken languages [7]. American Sign Language (ASL) and British Sign Language (BSL) are mutually unintelligible despite both being used in English-speaking countries. Meaning is conveyed not only through hand shapes but also through facial expressions, mouth patterns, and the location of the hands relative to the body [7]. Facial grammar plays a critical role: raised eyebrows indicate a yes/no question, furrowed eyebrows indicate a wh-question, and mouth movements can modify the meaning of a sign entirely [9]. This linguistic richness means that a system tracking only hand position captures an incomplete picture of the language, and a model trained on one sign language cannot be directly applied to another [1].

### B. From Gloves to Landmarks: Hardware Evolution

Early sign language recognition systems relied on instrumented gloves measuring finger bend angles and wrist orientation [10]. While achieving reasonable accuracy for isolated signs, these systems were impractical: expensive, uncomfortable, and capturing no facial or body information that is linguistically significant [11]. Depth cameras such as the Microsoft Kinect marked a step forward by enabling markerless skeleton tracking [12], but remained specialist hardware with significant cost barriers. Both approaches shared a fundamental limitation: they required hardware beyond a standard camera. The landscape changed significantly with MediaPipe [5], an open-source framework that performs real-time extraction of 21 hand landmarks, 33 pose landmarks, and 468 face landmarks directly from standard RGB camera input. Landmark representations are invariant to background, lighting, and clothing, meaning a model trained in one environment generalises far better to another a property critical for real-world deployment.

### C. Sequential Architectures for Sign Recognition

Sign language recognition is fundamentally a sequential problem: the meaning of a sign is conveyed through the trajectory of hand movements over time. Long Short-Term Memory networks [13] were among the first architectures to effectively model long-range sequential dependencies, using gating mechanisms to selectively retain or discard information. Bidirectional LSTMs [6] further enhance this by processing sequences in both forward and backward directions, leveraging future context particularly valuable since the beginning and end positions of a sign often carry more discriminative information than intermediate motion. Transformer architectures [14] offer an alternative via self-attention mechanisms that compute relationships between all positions simultaneously, but are data-hungry and can be outperformed by simpler recurrent architectures on smaller datasets [9] a trade-off directly observed during VoiceForAll's development.

### D. The PopSign Dataset and Feature Selection

The most significant dataset for isolated ASL recognition is the PopSign ASL v1.0 dataset, introduced by Starner et al. [1]. PopSign contains recordings of 250 ASL signs performed by 47 deaf signers using smartphones, totalling approximately 217,866 videos. A critical distinction from earlier datasets such as WLASL [8] is that the signers are deaf themselves, not hearing actors attempting to replicate signs, making the signing style authentic and representative of the natural variation found in the deaf community. The accompanying paper trained a Bidirectional LSTM [6] on 63 hand-only features, specifically 21 landmarks per hand multiplied by three coordinates ($x$, $y$, $z$), deliberately excluding pose and face landmarks, and applied left-hand mirroring to normalise all signers to appear right-handed [1]. This methodology achieved 84.2% accuracy on 250 signs using a signer-independent train/test split, meaning the signers in the test set were entirely different individuals from those in the training set, a significantly harder and more meaningful evaluation than a random split [1]. The PopSign paper serves as the primary benchmark and architectural blueprint for VoiceForAll, and understanding its methodology in detail was essential to diagnosing why earlier versions of the system performed poorly in real-world conditions despite achieving comparable test accuracy. The dataset itself, containing 248 of the 250 signs after preprocessing, forms the entire training corpus for the final version of VoiceForAll.

## III. System Design and Methodology

### A. Iterative Development Overview

The development of VoiceForAll followed an iterative, hypothesis-driven methodology across six distinct versions, each motivated by a specific question about what was limiting performance. The original ambition included continuous sign segmentation and cross-linguistic ASL-to-BSL transfer learning, but continuous segmentation remains an unsolved problem for single-camera landmark-based systems [9], and BSL transfer learning requires a target language dataset of sufficient size that does not currently exist publicly [3]. The scope was deliberately narrowed to isolated sign classification a significantly more tractable problem with a language model responsible for assembling recognised signs into coherent sentences [19]. Each subsequent version tested a specific hypothesis, with every decision informed by the results of the previous version.

TABLE I
Summary of VoiceForAll Development Versions

| Ver. | Architecture | Dataset | Test Acc. | Real-World |
|---|---|---|---|---|
| V1 | LSTM | WLASL-20 | 96%* | 40–47% |
| V2 | Transformer | WLASL-20 | 81% | inconsistent |
| V3 | LSTM | PopSign-13 | 72.80% | 54% |
| V4 | BiLSTM | PopSign-42 | 72.63% | 19% |
| V5 | Enh. Transformer | PopSign-42 | 70.80% | 24% |
| V6 | BiLSTM | PopSign-248 | **72.82%** | **87.5%** |

*V1 inflated due to data leakage

### B. Version 1 — Regular LSTM Baseline

Version 1 established the foundational pipeline: videos from the WLASL-2000 dataset [8] were processed through MediaPipe Holistic [5] to extract 225 landmarks per frame across 40-frame sequences, and a standard LSTM [13] was trained on 20 signs. The reported test accuracy of 96% appeared promising but live webcam performance reached only 40–47%, a gap traced to a data leakage bug where augmented copies of training videos appeared in the test set. Despite this, V1 validated the end-to-end pipeline and established that robust evaluation methodology was as important as model architecture.

### C. Version 2 — Transformer Architecture

Version 2 fixed the data leakage and introduced a Transformer architecture [14] with the dataset and feature set unchanged, making architecture the only variable. The Transformer achieved 81% honest test accuracy compared to a 57% LSTM baseline a substantial improvement. However, live webcam performance remained inconsistent, suggesting the bottleneck was not the model architecture but the 225-feature input representation, where pose landmarks were introducing noise from body movements unrelated to the sign.

### D. Version 3 — PopSign Dataset

Version 3 shifted focus from architecture to dataset quality, replacing WLASL with the PopSign ASL v1.0 dataset [1]. Training on a 13-sign subset provided 5,861 videos recorded by 47 verified deaf signers, compared to approximately 400 in earlier versions. Test accuracy reached 72.80%, but real-world testing revealed only 7 of 13 signs working reliably (54% live accuracy). The persistent gap indicated the problem was not solely data quality pose landmarks capturing torso and shoulder movement were introducing signer-specific noise that hurt generalisation.

### E. Version 4 — Bidirectional LSTM

Version 4 adopted the Bidirectional LSTM architecture [6] from the PopSign paper, simultaneously expanding the vocabulary from 13 to 42 signs. The 225-feature representation was retained so that architecture remained the only variable.

Test accuracy of 72.63% remained remarkably close to V3's 72.80% despite the task being three times harder, confirming the BiLSTM was genuinely stronger. However, real-world performance revealed the most severe gap of the entire project: only 8 of 42 signs worked reliably, giving a live accuracy of approximately 19% a 53 percentage point gap between laboratory and real-world performance.

### *F. Version 5 — Enhanced Transformer*

Version 5 tested whether a more sophisticated attention-based architecture could overcome the real-world gap. The Enhanced Transformer combined a multi-scale CNN feature extractor, two Transformer encoder blocks with 8 attention heads each, stronger regularisation including Gaussian noise and label smoothing, and combined global pooling [14]. Despite this substantial increase in complexity, test accuracy decreased slightly to 70.80% while real-world performance improved only marginally to 24%. The conclusion was unambiguous: attention mechanisms could not overcome noisy input features. Even the most complex model in the project failed to bridge the laboratory-to-real-world gap, pointing to one clear hypothesis for V6: the problem was not insufficient model complexity but insufficient feature quality.

### *G. Version 6 — Hands-Only Features*

Version 6 represented a fundamental change in philosophy: rather than increasing model complexity, the focus shifted entirely to input feature quality [15]. The preprocessing pipeline was extended to process all 248 available signs from the PopSign dataset [1], a total of 110,540 training videos, but with one critical modification: pose landmarks were removed entirely [5], reducing the feature vector from 225 to 63 features representing only the 21 hand landmarks multiplied by three coordinates. To validate the methodology before committing to the full dataset, training was performed incrementally across four stages, 16 signs, 43 signs, 82 signs, and finally 248 signs, confirming at each stage that the hands-only approach generalised before scaling further. The model architecture directly replicated the PopSign paper's BiLSTM configuration [1]: two stacked Bidirectional LSTM layers [6] of 128 units each, separated by `Dropout(0.5)`, followed by a softmax classification head, deliberately simpler than the Enhanced Transformer [14] tested in V5. Left-hand mirroring was implemented at inference time in the webcam scripts by inverting the $x$-coordinate of detected left-hand landmarks ($x \to 1 - x$), normalising all signs to appear right-handed regardless of the user's dominant hand [1]. Training on the University Tesla T4 GPU took approximately 3–5 hours, running for around 20–30 epochs before early stopping triggered, achieving a final test accuracy of 72.82%. The improvement in real-world performance was immediately obvious during webcam testing of the 16-sign model: where earlier versions had produced unstable, low-confidence predictions, V6 recognised most signs correctly on the first attempt with confidence scores frequently reaching 95–100%. Testing across all 16 signs confirmed 14 working reliably with an 87.5% real-world accuracy compared to 19–24% in V4 and V5. The most significant finding of the entire project emerged from this result: a simpler BiLSTM model [6] with clean hand-only features dramatically outperformed a complex Transformer [14] with attention mechanisms, CNN backbone, and extensive regularisation. Feature quality mattered far more than model complexity, a conclusion that directly validated the PopSign paper's methodology [1] and provides a clear lesson for future sign language recognition work.

### *H. System Pipeline*

The complete VoiceForAll pipeline integrates four components in sequence, transforming a hand gesture captured by a standard webcam into spoken English audio. The process begins with MediaPipe Holistic [5] running continuously on the webcam feed, extracting 21 hand landmarks per frame in real time. Rather than attempting automatic sign boundary detection, which remains an unsolved problem for single-camera systems [9], the system uses a manual trigger mechanism: the user performs a sign and presses `SPACE`, at which point the system records 60 frames, extracts the hand landmarks for each frame, assembles the sequence into a 63-feature input vector, and passes it to the BiLSTM model [6] for classification. The predicted sign and its confidence score are displayed on screen and added to a growing list of recognised words. The user repeats this process for each word they wish to communicate, then presses `ENTER` when ready to form a sentence. At this point the collected word list is passed to Llama 3.2 running locally via Ollama [19], which generates a grammatically correct English sentence, for example, converting the raw sign sequence `[``you'', ``help'', ``me'']` into *"Can you help me?"* The total latency from the final sign to spoken audio is approximately 1–2 seconds: BiLSTM prediction takes around 0.1 seconds, Ollama sentence generation takes 0.7–1.2 seconds, and text-to-speech startup adds approximately 0.2 seconds. Speech output is produced using the native macOS speech engine called directly from Python via `os.system(``say'')`, requiring no additional dependencies and working entirely offline [19]. A fallback mode activates automatically when Ollama is not running, joining the predicted signs directly as words and passing them to text-to-speech without grammatical correction, ensuring the system always produces output regardless of whether the LLM is available.

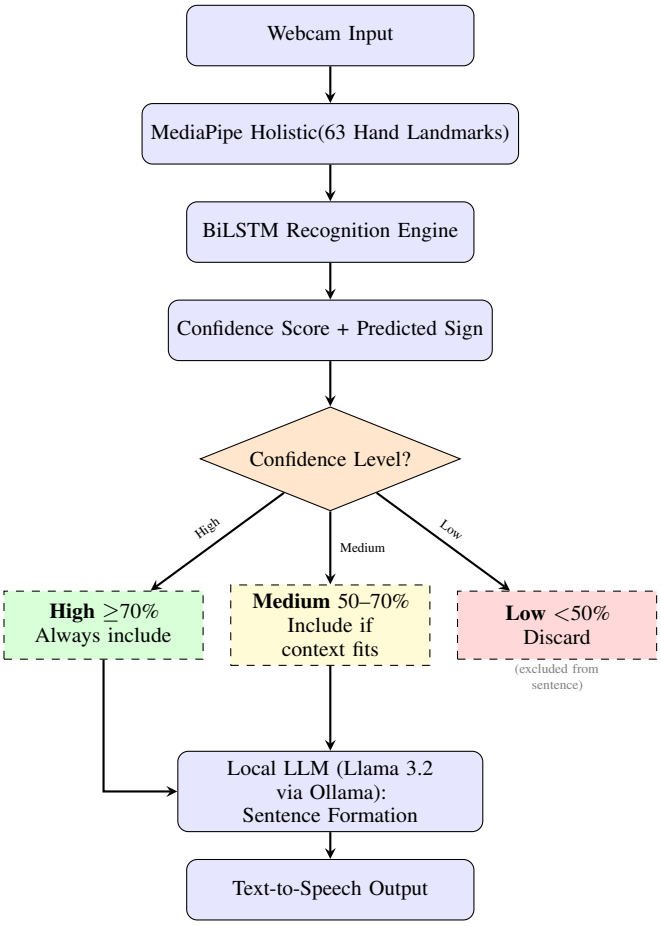


Fig. 1. VoiceForAll system pipeline. Each recognised sign is assigned a confidence score by the BiLSTM and routed through a three-tier filtering system before sentence formation. Low-confidence predictions are discarded; high and medium confidence signs are passed to the local LLM with contextual weighting. The entire pipeline runs offline with no internet connection required.

### I. Confidence-Based LLM Filtering

A key component of the VoiceForAll pipeline is the confidence-based filtering system that governs how recognised signs are converted into natural English sentences. Every sign prediction produced by the BiLSTM model [6] is accompanied by a softmax confidence score, and rather than simply concatenating recognised words in order, the system passes the full word list alongside their confidence scores to Llama 3.2 running locally via Ollama [19]. Words are categorised into three tiers: high confidence (70% and above), which the LLM is instructed to always include; medium confidence (50–70%), which are included where they fit the sentence context; and low confidence (below 50%), which are only used if they make sense given the surrounding words. The prompt explicitly instructs the model to construct one natural English sentence, add grammatical words such as articles and prepositions, and return only the sentence without explanation. This approach means the system produces grammatically correct, natural English output even when individual sign predictions are imperfect, a significant usability improvement over simple word concatenation [19]. If Ollama is not running, the system automatically falls back to joining the predicted words directly and passing them to text-to-speech, ensuring the system always produces spoken output regardless of whether the LLM is available. The minimum confidence threshold for any word to be included at all is 30%, below which predictions are considered too unreliable to contribute meaningfully to sentence formation.

TABLE II
WORKED EXAMPLE: CONFIDENCE-BASED FILTERING AND SENTENCE FORMATION

| Sign | Conf. | Tier | LLM Instruction |
|---|---|---|---|
| *you* | 94% | High (≥70%) | Always include |
| *help* | 62% | Medium (50–70%) | Include if context fits |
| *please* | 38% | Low (<50%) | Use only if makes sense |
| **Input to LLM:** ["you" 94%, "help" 62%, "please" 38%] | | | |
| **LLM output:** *"Can you help me please?"* | | | |
| **Spoken aloud via TTS:** *Can you help me please?* | | | |

## IV. EVALUATION

### A. Evaluation Approach

The system was evaluated using two complementary methods: quantitative test accuracy on a held-out dataset, and qualitative real-world webcam testing. A consistent 80/20 stratified train/test split with `random_state=42` was used across all six versions, ensuring results were directly comparable and any change in accuracy could be attributed to architectural or feature changes rather than variation in the data split. Real-world evaluation was conducted by testing every sign in the 248-sign vocabulary [1] through the live webcam application, producing a practical reliability rating for each sign. It is important to note that primary real-world testing was conducted by the project author, who is not a native signer, an acknowledged limitation addressed through expert evaluation with Professor Adam Schembri, discussed in Section 4.5.

### B. Test Accuracy and the Test-to-Real-World Gap

Test accuracy remained stable at 70–73% across V3–V6 despite substantial changes in architecture and vocabulary size, as shown in Figure 4. This stability is itself a significant finding: neither switching from LSTM to BiLSTM, nor adding CNN feature extraction, multi-head attention, and stronger regularisation, produced a meaningful improvement. The final V6 248-sign model achieved 72.82%, virtually identical to V3 despite the vocabulary being nineteen times larger.

The most revealing metric was the relationship between test accuracy and real-world webcam performance. Versions 3, 4, and 5 all achieved test accuracy in the 70–73% range, yet real-world performance told a dramatically different story (Table III): V3 achieved 54% live reliability, V4 collapsed to 19%, and V5 recovered only marginally to 24%. The gap reached as high as 53 percentage points in V4. The root cause was identified as the 225-feature input representation [5]: pose landmarks capturing shoulder position and body posture varied systematically between the controlled training data and live webcam input from a new user, causing the model to learn signer-specific characteristics rather than sign-specific ones [1].

Version 6 eliminated this problem by removing pose landmarks entirely, and the effect was immediate: real-world reliability on the 16-sign validation model reached 87.5%, exceeding the test accuracy of 81.27%, the first version where real-world performance exceeded laboratory performance.

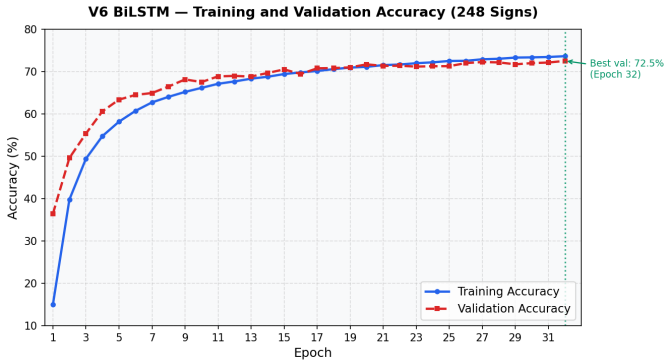


Fig. 2. Training and validation accuracy across epochs for the V6 BiLSTM model [6] on 248 signs [1]. The close tracking between training and validation accuracy confirms the model generalised rather than overfit, with early stopping triggering at epoch 32.

### C. Real-World Performance Across 248 Signs

Following the success of the 16-sign validation model, the complete 248-sign model was evaluated exhaustively. Across all 248 signs, 143 worked reliably (57.7%), 11 were marginal (4.4%), 4 worked occasionally (1.6%), and 90 failed entirely (36.3%).

TABLE III
REAL-WORLD PERFORMANCE ACROSS 248 SIGNS (V6 MODEL)

| Category | Signs | Percentage |
|---|---|---|
| Worked reliably | 143 | 57.7% |
| Marginal / inconsistent | 11 | 4.4% |
| Worked occasionally | 4 | 1.6% |
| Failed entirely | 90 | 36.3% |
| **Total** | **248** | **100%** |

Avg. confidence among working signs: 93.3%
41 signs achieved 100% confidence consistently

Among the 143 reliably working signs, the average confidence score was 93.3%, meaning the system is decisive when it recognises a sign correctly rather than producing marginal predictions. Forty-one signs achieved 100% confidence consistently. The overall live accuracy of 65.4% reflects evaluation by a non-native signer and should be understood as a lower bound, as discussed in Section 4.5, the system performed noticeably better when tested by an expert signer.

### D. Feature Analysis

Analysis of the 90 signs that failed to work reliably reveals three distinct failure categories, each with a different underlying cause and a different implication for future work. The first category, approximately 13 signs, fails due to a fundamental limitation of the hands-only feature design [5]. Signs such as *tooth*, *nose*, *chin*, *eye*, *ear*, *thirsty*, and *shh* are distinguished in ASL not by hand shape or movement but by the location of the hand relative to the face [7]. Pointing towards the nose versus pointing towards the tooth involves an identical hand shape in an almost identical position, with the only difference being which facial feature the fingertip is directed towards. Since VoiceForAll extracts only hand landmark coordinates and not facial position [5], these signs are indistinguishable to the model regardless of training data volume or architectural complexity. This is an honest and transparent limitation of the 63-feature design: the deliberate decision to use hands-only features, which eliminated the test-to-real-world gap and produced 87.5% real-world accuracy on visually distinct signs, simultaneously made it impossible to distinguish signs whose identity depends on facial location. Addressing this category would require reintroducing a small set of facial landmarks, specifically the positions of the nose, mouth, chin, and eyes, without reintroducing the full pose noise that caused the gap in earlier versions [5].

The second category, approximately 15 signs, fails due to visual similarity between hand shapes [15]. Pairs such as *bug* and *uncle*, *bye* and *bed*, *dog* and *later*, *wet* and *sleep*, and *yuck* and *uncle* involve hand movements that are genuinely almost identical, differing only in subtle details of wrist rotation, movement speed, or spatial position that are difficult to capture reliably from a single 2D webcam. These confusions are consistent with findings in the sign language recognition literature: visual similarity between signs is the primary source of classification error across all landmark-based recognition systems [9], and resolving them requires either significantly more diverse training data [1] or depth sensing hardware that captures 3D hand position [12].

The third and largest category, approximately 62 signs, represents a mixture of insufficient training data for rarer signs [1], signing precision issues where the sign as performed during testing differed subtly from the training data distribution, and residual domain shift between the smartphone-recorded training videos and the live webcam environment [1]. Many signs in this category worked occasionally or with careful execution, suggesting the model has learned a partial representation that could be strengthened with additional training data or more consistent signing technique.

### E. Confusion Matrix Analysis

Figure 3 presents the focused confusion matrix showing signs involved in genuine bidirectional confusions. Three distinct failure patterns are visible. The first confirms the visual similarity limitation [15]: *wake* and *awake* produced 85 total confusions, the highest of any pair, differing only in subtle timing and starting hand position. The second confirms the hands-only feature limitation [5]: *lips* and *mouth* produced 46 bidirectional confusions, and *nose* and *tooth* produced 25, as both pairs are distinguished purely by facial location which the hands-only model cannot detect. The third pattern is the *wolf* sign acting as a systematic default prediction for uncertain inputs, predicted 2,712 times but correct only 81 times, a

97.0% false positive rate caused by 62.3% zero-padded frames in its training data [1], a dataset quality issue rather than a fundamental model limitation.

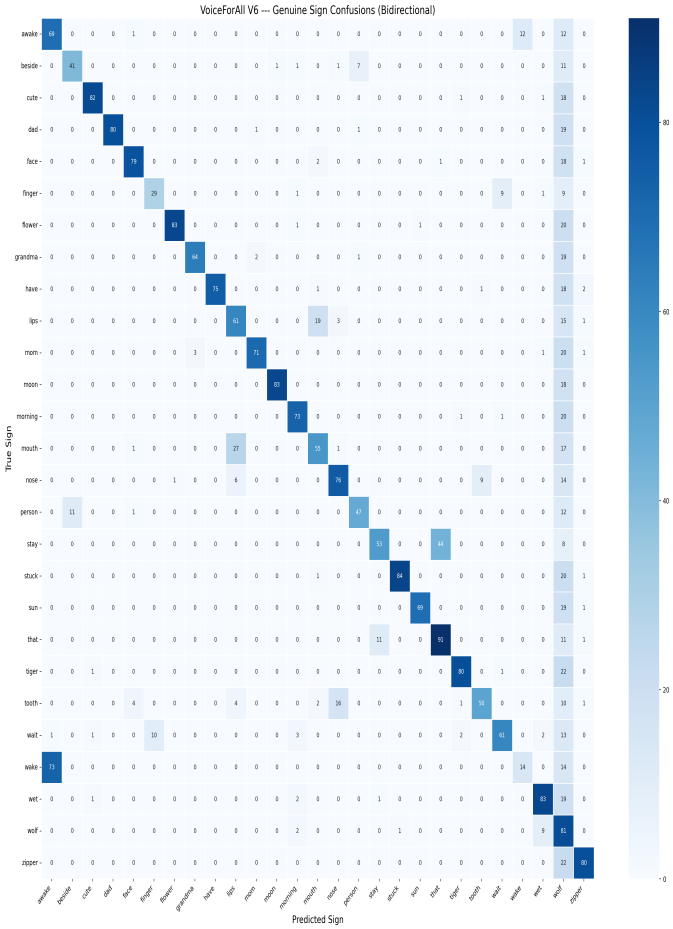


Fig. 3. Focused confusion matrix showing signs involved in genuine bidirectional confusions in the V6 248-sign model [6]. Three failure patterns are visible: visually similar hand shapes (*wake/awake*, *stay/that*) [15], face-location-dependent signs (*lips/mouth*, *nose/tooth*) [5], and the *wolf* garbage-collector pattern [1].

The complete $248 \times 248$ confusion matrix is available in the project GitLab repository. At vocabulary scale, the dominant diagonal confirms overall 72.82% test accuracy, with off-diagonal mass concentrated in three regions corresponding to the failure categories analysed above: face-location-dependent sign pairs in the upper-mid quadrant, visually similar sign pairs scattered along the near-diagonal, and the *wolf* garbage-collector column visible as a vertical stripe in the rightmost section of the matrix.

### F. Expert Evaluation

To provide a human-centred evaluation of the system beyond quantitative accuracy metrics, outreach was made to six organisations: deafPLUS Birmingham, the Royal Association for Deaf People, the British Deaf Association [3], RNID, University of Birmingham Disability Services, and BID Services Birmingham. The British Deaf Association responded by introducing the project to Professor Adam Schembri, a Professor of Sign Language Studies at the University of Birmingham and a world-leading scholar in the field, who agreed to review and test the system in person.

A live demonstration was conducted in which Professor Schembri tested VoiceForAll himself, performing signs directly into the webcam and observing the system's recognition and sentence formation pipeline in real time. The system performed noticeably better during his testing than during the author's own evaluation, a result Professor Schembri attributed to his familiarity with correct ASL sign forms. This finding directly supports the hypothesis that the 65.4% overall real-world accuracy reported in Section IV-C underestimates the system's true capability with accurate signing: the system responds well when signs are performed correctly, and the gap between the author's evaluation and Professor Schembri's testing reflects differences in signing precision rather than fundamental limitations of the recognition pipeline.

Professor Schembri provided structured qualitative feedback on the system. He confirmed that the core sign-to-speech pipeline functions effectively and that a tool of this kind has genuine communicative utility. He identified bidirectional communication and facial expression recognition as the most important directions for future development, observations that directly align with the technical limitations identified in the failure analysis in Section IV-D. He also provided valuable context on the deaf community's perspective on sign language technology, noting that community acceptance is significantly influenced by whether technology is perceived as empowering deaf individuals rather than attempting to fix them, a consideration that informed the privacy-preserving, offline design of VoiceForAll and its focus on giving deaf signers a voice rather than requiring hearing people to learn nothing.

This expert evaluation provides qualitative evidence that complements the quantitative results presented in this section. A system validated by a world-leading sign language scholar who tested it directly and confirmed its communicative utility represents a meaningful real-world endorsement that goes beyond laboratory accuracy metrics. Broader community evaluation with native deaf signers remains an important direction for future work, and the methodology outlined during outreach, asking participants to convey a real message of their choosing rather than performing a fixed sign list, provides a clear framework for that evaluation.

### G. Comparison with the PopSign Paper

The final V6 model achieved 72.82% test accuracy compared to the PopSign paper's benchmark of 84.20% [1], a gap of 11.38 percentage points. Two methodological differences account for the majority of this gap. First, the PopSign paper employed a signer-independent evaluation where different individuals appeared exclusively in training and test sets, a significantly harder evaluation than the random 80/20 split used in VoiceForAll. Second, left-hand mirroring was applied at inference time but not consistently during training preprocessing, introducing additional variation the model needed to

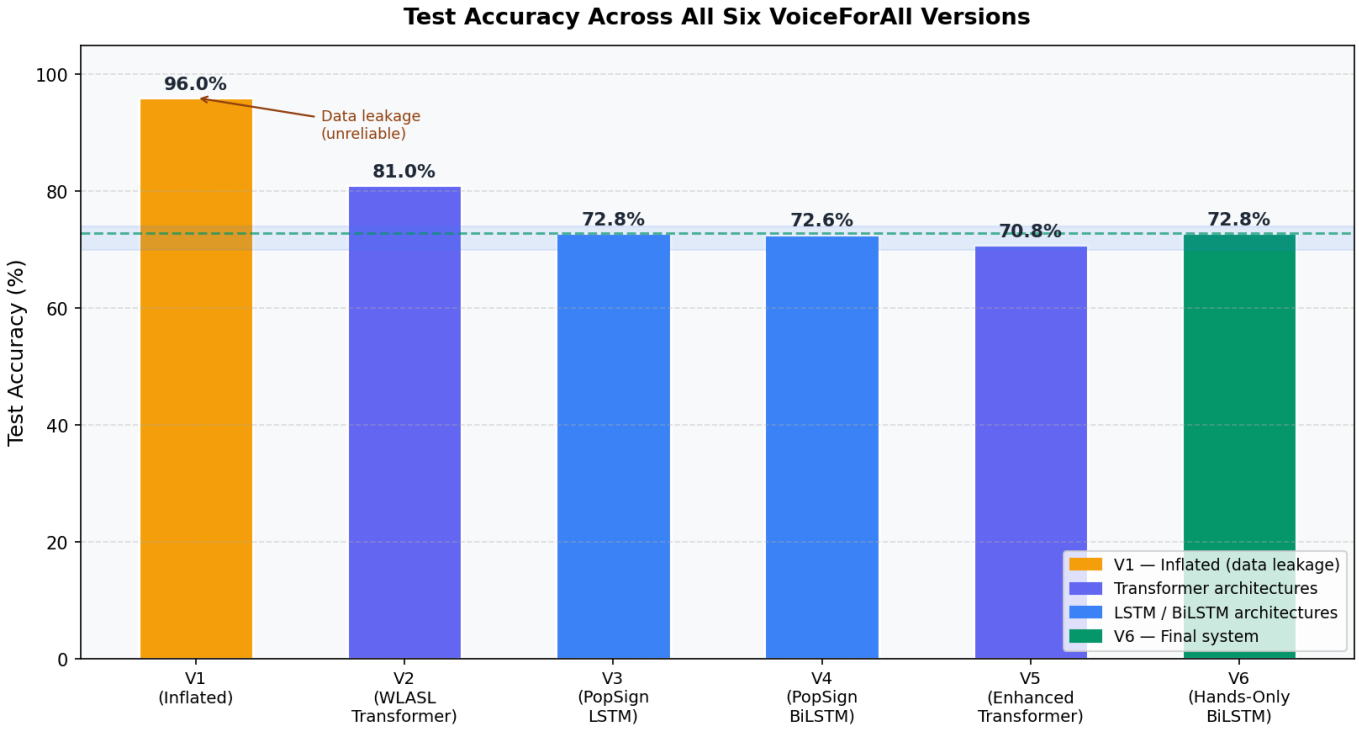


Fig. 4. Test accuracy across all six VoiceForAll versions. The stability band (70–73%) across V3–V6 demonstrates that architectural changes produced no meaningful improvement, directly supporting the finding that feature quality matters more than model complexity [15]. *Note: V1 accuracy is unreliable due to data leakage [8]. V2 used the WLASL dataset [8] and is not directly comparable to V3–V6, which all use PopSign [1].*

learn. Both limitations are well understood and addressable in future work. Importantly, the real-world failures observed are more convincingly explained by sign similarity and the hands-only feature limitation than by signer-specific overfitting [5], and the expert evaluation with Professor Schembri provides practical evidence that the system generalises meaningfully to new users when signs are performed correctly.

## V. Critical Analysis and Limitations

### A. Hands-Only Feature Trade-Off

The decision to use only 63 hand landmarks [5] was the single most impactful design choice in the project, producing the largest improvement in real-world performance across all six versions [15]. However, it introduced a fundamental limitation that cannot be resolved through additional training data or architectural improvements alone. Signs distinguished in ASL by the location of the hand relative to the face, such as *tooth*, *nose*, *chin*, and *mouth*, are inherently indistinguishable to a hands-only system [7]. This is not a failure of the model to learn correctly; it is a consequence of providing insufficient information to make the distinction [5]. This limitation is rooted in the linguistic structure of ASL itself: as established by Stokoe [7], the location of a sign relative to the body and face is a primary phonological parameter, as linguistically significant as handshape or movement. The trade-off is clear and deliberate: hands-only features eliminated the signer-specific pose noise that caused the catastrophic test-to-real-world gap in V3–V5, producing a system that generalises well to new users. The most promising solution is not to reintroduce full pose landmarks but to selectively add a small number of facial reference landmarks, the nose, mouth, chin, and eyes, providing spatial context without reintroducing irrelevant body pose information [15].

TABLE IV
Failure Analysis: Three Categories of Unrecognised Signs

| Category | Signs | Root Cause |
|---|---|---|
| Face-location dependent | ∼13 | Hand location relative to face not captured by hands-only model |
| Visually similar | ∼15 | Nearly identical hand shapes differing only in subtle wrist rotation or speed |
| Data / domain issues | ∼62 | Insufficient training data, signing precision, or domain shift |

Total failing signs: 90 of 248 (36.3%)

### B. Methodological Limitations

Two methodological differences from the PopSign paper [1] warrant acknowledgement. First, VoiceForAll uses a random 80/20 stratified split rather than a signer-independent evaluation, meaning the same 47 signers appear in both training and test sets. This makes the 72.82% figure methodologically distinct from the paper's 84.20% benchmark, the two numbers measure different things and are not directly comparable. However, real-world failures are more convincingly explained by sign

similarity and the hands-only limitation than by signer-specific overfitting, and the expert evaluation with Professor Schembri, an entirely new signer never seen during training, produced noticeably stronger results, providing practical evidence that the system generalises to new users [1]. Second, left-hand mirroring was applied at inference time but not consistently during training preprocessing, introducing variation the model needed to learn. Both limitations are well understood and addressable in future work.

### C. Single User Evaluation and Platform Dependency

Real-world testing was conducted primarily by the project author, who is not a native signer. Signs were learned from reference videos rather than acquired through years of natural signing experience, meaning signs categorised as unreliable may reflect execution quality rather than system capability [1]. The expert evaluation with Professor Schembri partially addresses this limitation, his testing produced stronger results, suggesting the 65.4% overall accuracy figure is a lower bound rather than a ceiling [3]. Additionally, the text-to-speech implementation uses the macOS native speech engine, meaning the system currently runs only on macOS. Replacing this with a cross-platform library such as `pyttsx3` would be a straightforward modification. The dependency on Ollama [19] being installed locally introduces a setup barrier that may be significant for the deaf community members the system is designed to serve, though the fallback mode ensures the system always produces spoken output regardless of whether the LLM is available.

### D. Language Limitation

The system currently supports American Sign Language only, a limitation of particular significance given the project was developed in the United Kingdom where BSL is the primary sign language [3]. The original ambition included cross-linguistic transfer learning from ASL to BSL, but this was scoped down due to the absence of a sufficiently large publicly available BSL dataset [1]. The hands-only feature representation used in V6 [5] is particularly well suited to transfer learning: hand shape and movement are more consistent across sign languages than body pose [7], making the learned representations a transferable foundation for BSL extension.

## VI. Conclusions and Future Work

### A. What Was Achieved

VoiceForAll demonstrates that a real-time, end-to-end sign language to speech system is achievable on a standard laptop with no specialist hardware, no internet connection, and no prior sign language knowledge from the hearing participant. The system recognises 248 ASL signs using a Bidirectional LSTM model [6] trained on 110,540 videos from the PopSign ASL v1.0 dataset [1], converts recognised signs into grammatically correct English sentences using a locally-running large language model [19], and speaks them aloud via text-to-speech, completing the full communication pipeline from hand gesture to spoken audio in approximately 1–2 seconds. Among reliably recognised signs, average confidence exceeds 93%, and 41 signs are recognised with 100% confidence consistently. The 16-sign validation model achieved 87.5% real-world accuracy, and expert evaluation by Professor Adam Schembri, a Professor of Sign Language Studies at the University of Birmingham, confirmed that the system functions effectively and has genuine communicative utility in real interactions.

The most important finding of the project is not a performance metric but a methodological insight: feature quality matters more than model complexity [15]. Across six versions testing regular LSTM [13], Bidirectional LSTM [6], and increasingly sophisticated Transformer architectures [14], no architectural change produced a meaningful improvement in real-world performance. The single change that transformed the system, removing 162 pose landmarks [5] and retaining only 63 hand landmarks, produced a larger improvement in real-world reliability than all architectural changes combined. This finding directly validates the methodology of the PopSign paper [1] and provides a clear, evidence-based lesson for future sign language recognition work: ensure that input features are sign-specific before optimising model architecture [15].

### B. From Ambition to Delivered System

The delivered system represents a significant but principled narrowing of the original scope. Three ambitious goals, continuous sign segmentation, bidirectional sign-to-speech and text-to-sign communication, and cross-linguistic ASL-to-BSL transfer learning, were each scoped down through honest assessment of technical feasibility within available data, computational resources, and time constraints. Continuous segmentation remains unsolved for single-camera systems [9], and BSL transfer learning requires target language data that does not currently exist publicly [3]. The scope discipline that produced the final system was not a compromise but a deliberate research decision: the iterative versioning methodology made possible by that discipline produced the project's central finding, that removing features improved performance more than adding architectural complexity [15], an insight that a single ambitious attempt would never have revealed.

### C. Future Work

Several concrete directions emerge from the findings and limitations of this project. The highest priority is broader community evaluation with native deaf signers, building on the expert evaluation with Professor Schembri and extending it to a wider range of participants from the UK deaf community [3]. The second priority is implementing a signer-independent train/test split [1], which would enable a more meaningful comparison with the PopSign paper benchmark. The third direction addresses the hands-only limitation directly: selectively reintroducing facial reference landmarks, the nose, mouth, chin, and eyes [7], would enable the system to distinguish face-location-dependent signs without reintroducing the pose noise that caused the test-to-real-world gap in earlier versions [5]. The fourth direction is BSL extension through transfer learning [6] from the ASL-trained model, the original ambition of the project and arguably its most impactful potential contribution.

The fifth direction is text-to-sign video playback using pre-recorded signing videos already available in the PopSign dataset [1], which would complete the bidirectional communication pipeline with no generative AI required. Finally, cross-platform deployment through replacing the macOS-specific text-to-speech component with a platform-independent library would make VoiceForAll accessible to a significantly wider audience, moving it closer to the genuinely accessible, universally available communication tool that motivated the project from the beginning [3].

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